# MULTI-SENSOR ANIMAL DETECTION AND AUTOMATED RESPONSE SYSTEM FOR CROP PROTECTION


**Group 18:**

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## INTRODUCTION &BACKGROUND

Wild animals such as elephants and wild pigs often cause serious damage to agricultural crops, especially in rural areas where continuous monitoring is difficult. Traditional protection methods such as fences or manual guarding are not always effective and require significant human effort.

With the advancement of sensor technology, automated systems can be developed to detect animal intrusion more accurately. By using multiple sensors, it is possible to identify animal presence, estimate their size and number, and activate deterrent mechanisms such as sound and light only when necessary.

This project proposes multi-sensor animal detection and deterrent system to provide a practical and low-cost solution for reducing crop damage while minimizing the need for continuous human supervision

## OBJECTIVES

**Detection & Identification**

- To detect animal entry into a predefined region using a PIR sensor.
- To estimate the size/type of animal using ultrasonic and IR beam sensor data.
- To estimate the number of animals entering the region using vibration sensor data patterns.

**Automated Response**

- To automatically activate deterrent mechanisms of light and sound based on detected animal characteristics.
- To vary the intensity of sound and light depending on the estimated size and number of animals.

**System Development**

- To design and implement a custom vibration sensor as part of the sensing system.
- To integrate sensors, actuators, and microcontroller into a functional prototype.

## AIM

- To design and develop an intelligent system that can detect and respond to wild animals approaching cultivated land.
- To reduce crop damage by providing an automated, real-time animal detection and deterrent mechanism.
- To create a low-cost and practical solution suitable for agricultural environments.

## SCOPE

- Detection of animal movement near a protected agricultural boundary.
- Differentiation between three animal size categories.
- Estimation of number of animals entering the monitored area.
- Activation of light and sound deterrent system automatically.
- Prototype-scale demonstration suitable for laboratory testing.

## LITERATURE REVIEW

- **SENSOR BASED WILD ANIMAL DETERRENT SYSTEMS**

1. In 2023 group of people in Spain developed a system to detect the presence of wild elephants and alert the landowner. With the help of infrared barriers at different heights they detected the presence of adult elephants and gave alert to the landowners. Not only adult elephants but also, they were able to detect Giraffe and unauthorized large vehicles entry as well.

Research gap:

- They used this system mainly to detect elephants, where they can detect different kinds of wild animals which cause harm for cultivation land.
- This system has been designed to give only alerts for landowners where they didn’t have any automatic chasing mechanisms.

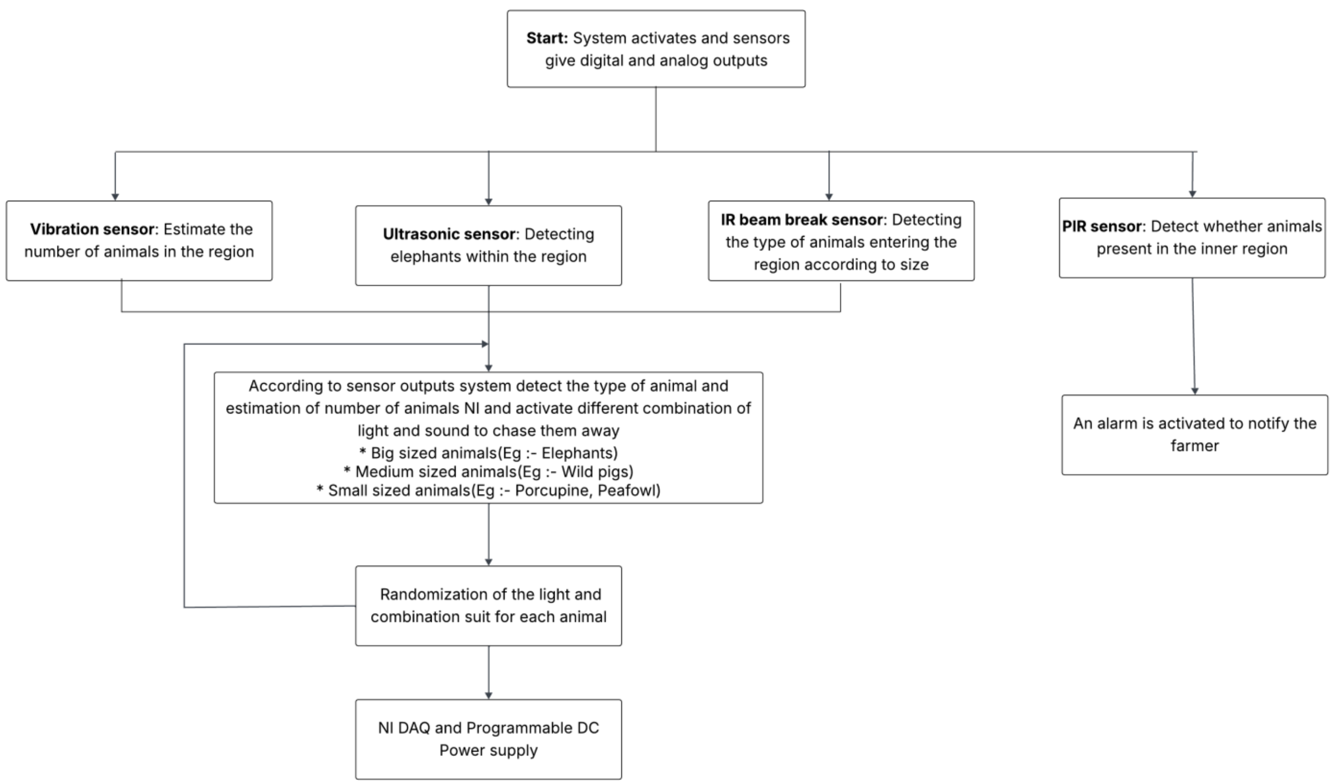
- **SENSOR AND AI INTEGRATED DETERENT SYSTEMS**

1. In 2024 a group of people in Taiwan developed a wild bird repellent system. They detected the presence of birds with the help of camera and AI model (YOLO). The trained model evaluates the image from camera with the data set it has and detects the specific type of wild bird. A rotatable laser control unit is used to chase the bird away from field.
2. In 2023 a team of IIT students in Sri Lanka developed a wild animal detection and deterrent system. With the help of audio, image processing and deep learning model they were able to detect the type of animal. Based on that the alarm system was triggered to alert the landowners.
3. In 2025 a group of people in Bhutan developed a smart system which detects the presence of seven different species of wild animals and alert the landowners with the help of night vision camera, ultrasonic sensor and deep learning YOLOv8 algorithm.

Research gap:

- These systems mainly depend on AI models. In rural village areas it is hard to get stable internet connections for the proper operation of AI modules.
- These AI integrated systems will be expensive where farmers couldn't afford it.

## METHODOLOGY




Here IR beam break sensor mark the outer boundary around the field and by using at different heights we can distinguish between three types of animals based on their size. Then using analog Vibration sensor output patterns, we can estimate the no.of animals present in the region. For better accuracy an ultrasonic sensor placed at a certain height, we can detect whether big sized animals like elephants present within the region. Based on output of these sensors based on type of animal and no.of animals, combination of lights and sound will be activated as the chasing mechanism

But if somehow animals approached to the cultivation field, PIR sensor is placed to identify those situations and farmer will receive an alert message to his phone.

## ESTIMATED BUDGET

| **Components** | **Estimated Cost** |
|---|---|
| Ultrasonic Sensor (HC-SR04) | LKR 240.00 |
| PIR Sensor (HC-SR501) | LKR 230.00 |
| Analog Vibration Sensor | LKR 690.00 |
| LEDs | LKR 100.00 |
| Buzzer | LKR 150.00 |
| Bread Board and wires | LKR 1000.00 |
| LM358 | LKR 20.00 |
| Resistors | LKR 100.00 |
| Phototransistor | LKR 1800.00 |
| Potentiometer | LKR 600.00 |
| IR LED | LKR 180.00 |
| ESP32 microcontroller | LKR 1790.00 |
| **Total** | **LKR 6900.00** |

## CONCLUSION

The proposed Animal Detection and Chasing System aims to provide a practical and efficient solution to reduce crop damage caused by animal intrusion. By using multiple sensors, the system can detect the presence of animals, estimate their size category and approximate number, and activate suitable deterrent mechanisms such as sound and light. Proposed solution is designed as a low-cost and easily implementable system that can support farmers and land owners by minimizing the need for continuous human supervision. The project demonstrates how sensor technology can be applied to solve real world problems effectively.

## REFERENCES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